

FSD Laboratory 02

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Aim: Design and develop a responsive web page using Bootstrap front end framework.

Objectives:

1. To understand HTML tags
2. To learn the styling of web pages using CSS
3. To learn Bootstrap Front End Framework.

Theory:

1. Bootstrap Grid System.

The Bootstrap Grid System is a flexible layout system used to create responsive web page designs. It is based on a 12-column structure and uses rows and columns to arrange content so that it automatically adjusts to different screen sizes such as mobile, tablet, and desktop.

Bootstrap divides the page into rows, and each row is divided into columns whose total width must add up to 12. Columns are created using predefined classes, which allow developers to control the layout across different devices.

The grid system is responsive and supports multiple screen sizes using breakpoints such as extra small, small, medium, large, and extra large screens. This ensures that the website layout adapts smoothly to various devices without writing complex CSS.

In summary, the Bootstrap Grid System simplifies responsive design by providing a structured, flexible, and

2. Bootstrap .container and .container-fluid class.

In Bootstrap, .container and .container-fluid are layout classes used to wrap content and align it properly within the grid system.

.container

The .container class provides a fixed-width layout that changes at different screen sizes. It centers the content and adds left and right padding. The width of the container adjusts according to predefined breakpoints, making it suitable for most standard website layouts.

`.container-fluid`

The `.container-fluid` class provides a full-width layout that spans the entire width of the viewport. It also includes padding but does not have fixed width limits. This class is mainly used when a layout needs to stretch across the full screen, such as dashboards or full-width sections.

FAQ:

1. What is a responsive website?

A responsive website is a website that automatically adjusts its layout, content, and design according to the screen size and device on which it is viewed, such as mobile phones, tablets, laptops, and desktops. The goal of a responsive website is to provide an optimal viewing and user experience across all devices.

Responsive websites use flexible layouts, images, and CSS media queries to adapt to different screen resolutions. This ensures that text remains readable, images scale properly, and navigation is easy to use without zooming or horizontal scrolling.

Responsive web design improves accessibility, usability, and performance. It is widely used in modern web development frameworks such as Bootstrap.

2. How Bootstrap helps to design a responsive website?

Bootstrap helps in designing a responsive website by providing a mobile-first framework with ready-made CSS and JavaScript components. It includes a responsive grid system that divides the page into 12 columns, allowing layouts to automatically adjust according to different screen sizes.

Bootstrap offers predefined classes for various device sizes, which makes it easy to control how content appears on mobile phones, tablets, and desktops without writing complex CSS. It also provides responsive utilities, flexible images, and navigation components that adapt smoothly to different screen resolutions.

By using Bootstrap's built-in features, developers can quickly create websites that are consistent, responsive, and compatible across devices and browsers using Bootstrap.

Output: Screenshots of the output to be attached.

Problem Statement:

Design and develop a responsive web page (For example student registration, course enrollment, library management system, online shopping system etc.) using Bootstrap front end framework. Web pages should contain HTML5 elements (Use all possible formatting for example font, colour etc.). Use all possible formatting for example,

- font,
- colour etc.

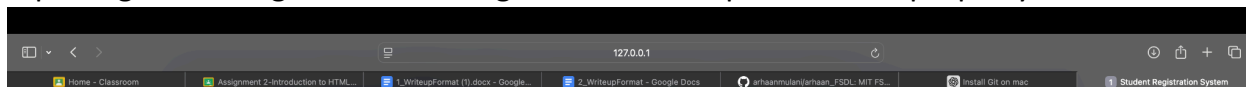
Web page should include various

- images,
- links within the page,
- tables
- lists
- form elements
- paragraphs
- links to other pages for navigation,
- new tabs.

Use Cascaded Style Sheets (CSS) to style the web pages designed. Make use of

- Three types of styles (Inline, Internal and External)
- border,
- margins,
- padding,
- navigation,
- dropdown list
- tag selector
- class selector
- id selector

- exploring background image and position property etc.



Student Registration Form

Name Enter name	Email Enter email
Gender Male	Course B.Tech
Address	
Submit	

Available Courses

Computer Science
Information Technology
Artificial Intelligence

Registered Students

ID	Name	Course
1	Arhaan	B.Tech
2	Rahul	MCA

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arhaan_FSDL Public

assignment2 had recent pushes 20 seconds ago

assignment2 2 Branches 0 Tags

This branch is 1 commit ahead of main

arhaanmulani Assignment 2: Added code folder 4a516d5 · now 3 Commits

FSDL_ASS2	Assignment 2: Added code folder	now
LICENSE	Initial commit	4 days ago
file1.txt	Added three text files to main branch	2 days ago
file2.txt	Added three text files to main branch	2 days ago
file3.txt	Added three text files to main branch	2 days ago

```
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git commit -m "Assignment 2: Added code folder"

[assignment2 4a516d5] Assignment 2: Added code folder
3 files changed, 211 insertions(+)
create mode 100644 FSDL_ASS2/about.html
create mode 100644 FSDL_ASS2/index.html
create mode 100644 FSDL_ASS2/styles.css
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git push origin assignment2

Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 14 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.33 KiB | 2.33 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'assignment2' on GitHub by visiting:
remote:   https://github.com/arhaanmulani/arhaan_FSDL/pull/new/assignment2
remote:
To https://github.com/arhaanmulani/arhaan_FSDL.git
* [new branch]      assignment2 -> assignment2
arhaan@Arhaans-MacBook-Pro arhaan_FSDL %
```

```
main
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % cd ~/Documents/arhaan_FSDL

arhaan@Arhaans-MacBook-Pro arhaan_FSDL % ls -a
.          .git          file2.txt    LICENSE
..         file1.txt    file3.txt
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git checkout main
git pull origin main

Switched to branch 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/arhaanmulani/arhaan_FSDL
* branch      main      -> FETCH_HEAD
Already up to date.
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git checkout assignment2

Switched to branch 'assignment2'
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git branch

* assignment2
main
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % /Users/arhaan/Documents/arhaan_FSDL
zsh: permission denied: /Users/arhaan/Documents/arhaan_FSDL
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % /Users/arhaan/MIT /LABORATORY/FSDL_ASS2
zsh: permission denied: /Users/arhaan/MIT /LABORATORY/FSDL_ASS2
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % mv "/Users/arhaan/MIT /LABORATORY/FSDL_ASS2" .

arhaan@Arhaans-MacBook-Pro arhaan_FSDL % ls

file1.txt      file2.txt      file3.txt      FSDL_ASS2      LICENSE
arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git add FSDL_ASS2

arhaan@Arhaans-MacBook-Pro arhaan_FSDL % git commit -m "Assignment 2: Added code folder"
```